Solar Filament Instance Segmentation with Native Detail and Auditable Model Selection

A Solution to the Solar Filament Segmentation Challenge 2026

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Abstract

We present a pixel-only system for instance segmentation of solar filaments in GONG H-α images. Four native-resolution ResNet34 U-Nets and four Mask R-CNN models provide complementary semantic and object evidence; a training-derived selector and two crop refiners preserve fine structure while separating instances. The accepted H9b no-observatory entry achieved public Panoptic Quality (PQ) **0.39**, Kaggle submission **58112557**, on 8 November 2026. Local temporal-block studies, uncertainty estimates, failure analyses and negative controls are reported separately from this official measurement. A subsequent audit reproduced all **1,241 submitted RLE rows byte for byte** from frozen H9b crops and the original refiners. A full 180-image replay from reconstructed stages retained 1,241 instances but matched 1,101 RLE rows exactly by position. This stage-coupling check does not establish a clean execution of the revised raw-JPEG notebook. The public release includes source, model weights, integrity manifests and executable notebooks. This second revision, with evidence through 13 November 2026, corrects the semantic-encoder description and reports the later cross-fold and reproducibility audits.

Keywords

solar filaments, instance segmentation, Panoptic Quality, reproducibility

1 Introduction

This report goes over the details of our submitted solution to the Solar Filament Segmentation Challenge 2026 [1]. The challenge, launched on July 10, 2026, is organized by the Earth-Space AI Research (ESAIR) Lab[1]. This challenge utilizes the MAGFiLO dataset for evaluation of the segmentation algorithms [2].

Thin, low-contrast structures make filament segmentation sensitive to both spatial resolution and instance grouping. A mask can recover dark pixels while splitting one physical filament, joining adjacent filaments, or retaining unrelated dark features. Our contribution is an integrated native-detail pipeline with explicitly recorded model selection, uncertainty and artifact provenance. We use established segmentation and detection families [5–8]; the release attributes the public native-resolution recipe credited in its source files. The reported checkpoints were trained for this entry. We do not claim a new general-purpose neural architecture.

[1] www.esairlab.com/

Kaggle Competition, kaggle.com/competitions/filament-segmentation-2026 2026.

2 Methodology

2.1 Data, scoring and temporal validation

The competition archive contains 707 labeled training observations, 180 test JPEGs, 1,154 training annotator–image records and 8,199 filament annotations. Images are 2048×2048 grayscale. The median annotation occupies 1,228 native pixels. Several experts can independently annotate the same physical observation. Consequently, multiple annotation records are not independent images and cannot be randomly split as if they were.

All annotators of a physical image are grouped before splitting. Four temporal blocks use a five-day exclusion buffer between training and held-out dates. Calibration selects thresholds and post-processing; paired evaluation uses reserved observations. Bootstrap uncertainty resamples physical images, preserving all their annotation records together. Some partitions have subsequently been examined repeatedly; later analyses of those partitions are explicitly exploratory. The exclusion buffer reduces short-range temporal similarity but does not prove complete independence of solar structures.

We follow the official self-evaluation notebook [3]. For each annotator–image record, the same prediction is compared with that annotator's instances. Every pair with IoU strictly greater than 0.5 contributes to the true-positive overlap sum; predictions and annotations without a qualifying match contribute FP and FN. Counts are pooled over records:

$$\mathrm{PQ} = \frac{\sum_{(g,p)\in TP} \mathrm{IoU}(g,p)}{|TP| + 0.5|FP| + 0.5|FN|}. \tag{1}$$

The local evaluator preserves the reference overlap-matrix semantics, including its treatment of qualifying pairs. It does not substitute an image-wise average of Dice. The evaluation concept follows panoptic segmentation [4], while competition comparability requires the host's concrete implementation.

At test inference, all features come from the supplied individual H-α image. The accepted no-site variant excludes observatory identities and date features. No external test annotations, adjacent frames or manually inferred test labels enter this predictor. Dates are useful for constructing training partitions, not for its inference decisions.

2.2 Semantic maps and instance hypotheses

Native semantic branch. H3 uses **four ResNet34-encoder U-Nets**, one for each F1–F4 training configuration. Each uses a one-channel input, a soft annotator-consensus foreground target, and binary cross-entropy plus soft Dice loss. Training uses 512×512 crops with rotations, reflections and intensity augmentation. The released

predictor normalizes by statistics within a fixed solar-disc region and evaluates the full native image with eight rotation/reflection views. View probabilities are averaged and quantized to 8-bit maps. The accepted H3 ensemble does not contain an EfficientNet semantic checkpoint; EfficientNetV2-S is used in the refinement stage described below.

Object branch and consensus. Four Mask R-CNN models provide object proposals. The published consensus operation preserves the original fold-major order of normal and unflipped horizontal views. It uses the frozen score gate 0.30, mask threshold > 127, cross-view support at mask IoU 0.50, an unsupported-score multiplier 0.75, stable confidence sorting and greedy mask non-maximum suppression at IoU 0.50. These values describe the frozen historical pipeline, rather than a new leaderboard-tuned search.

The H6 stage fuses semantic and detector evidence and produces candidate components. A selector fitted on competition TRAIN predicts which candidates to retain. Its no-site version excludes date and observatory features. Candidate selection reflects the marginal PQ cost of false positives: for an isolated additional proposal with match probability p and expected matched overlap u, the approximation $pu > \mathrm{PQ}/2$ gives a useful operating-point diagnostic. At PQ 0.39 and $u = 0.64$, the implied break-even probability is about 0.30. This approximation motivates caution with unsupported detections; it is not a replacement for empirical calibration.

2.3 Native-detail refinement and mask assembly

H9b applies two U-Net-style crop refiners: a ResNet34 model and an EfficientNetV2-S model. Rotated 128×384 crops align each candidate's long axis. The four input channels are flat-field image intensity, fused probability, the candidate seed, and other candidates. Each refiner therefore has image appearance and explicit instance context. The published affine geometry maps refined probabilities back into the original image.

Instances are assembled in a fixed confidence order with exclusive pixel ownership. The output is a 2048×2048 mask per predicted filament, encoded using compressed COCO RLE in Fortran order. Identifiers, row ordering, geometry and exact CSV bytes are checked independently. The submitted artifact has 1,241 rows and SHA-256 beginning `24b3ea1c8e151c30`; the full digest is in the public provenance record [9]. A differently named historical intermediate has a different hash and is not used as proof of the accepted upload.

3 Evaluation

3.1 Official and local measurements

Table 1 records historical public Kaggle results. The strongest accepted entry is the pixel-only H9b variant. Public scores are rounded to two decimal places on approximately half of the competition test set. Local PQ uses labeled training partitions and must be interpreted separately; neither local confidence intervals nor numerical replay tests are new official scores.

On 567 purged held-out training observations, the H6 control obtained native PQ 0.4465. H9 reached 0.4491, a paired difference of +0.0026 with a 95% interval [+0.0002,+0.0051], positive in three of four blocks. The two-refiner H9b variant reached 0.4502, a difference of +0.0037 [+0.0010,+0.0064], positive in all four blocks.

Table 1: Historical public leaderboard measurements.

Entry	PQ	Date (2026)
Four-block Mask R-CNN (P1)	0.32	6 Nov
NeuroPixel multiscale ensemble	0.35	28 Sep
Three-family P9	0.36	6 Nov
Three-family P12	0.37	7 Nov
H6 learned selector	0.38	7 Nov
H9b no-site, ID 58112557	**0.39**	**8 Nov**

Removing observatory features yielded +0.0030 [approximately +0.00001,+0.0059], positive in three blocks.

The preregistered local promotion criterion required a positive effect in at least three blocks, average paired improvement above +0.005, and a strictly positive 95% interval. The H9b exploratory cross-block improvement did not satisfy the effect-size threshold. Its higher public score is nevertheless an observed historical result. We preserve both facts instead of treating the leaderboard increase as proof that the local promotion rule passed.

3.2 Morphology and uncertainty

The H6 diagnostic retained-set recall was 0.622 over 6,327 reference instances. Recall was 0.379 below 500 pixels, 0.632 at 500–1,000, 0.686 at 1,000–2,500, 0.685 at 2,500–7,000, and 0.594 above 7,000 pixels. The matched-object median IoU was 0.689, while median whole-foreground Dice per annotator record was 0.700. Approximately 11.2% of reference instances had best IoU in $(0.3, 0.5]$, identifying plausible refinement targets without using hidden test labels.

Our supplementary fragmentation diagnostic flagged 3.94% of reference instances with at least two predictions lying at least halfway inside that instance. The merging diagnostic flagged 97 of 5,708 prediction–record appearances as covering at least 20% of two or more reference instances. These coverage-based definitions are stated explicitly; Figure 2 reports matched IoU and record-wise Dice, rather than claiming an unmeasured distribution. Annotation disagreement and incomplete correspondence between darkness and filament identity remain substantial error sources.

3.3 Controlled ablations and subsequent checks

The P58 small-filament sampling study completed 3,600 steps per arm on one RTX 3090. On its fixed F1 calibration cohort (117 observations, 190 annotation records), treatment PQ was 0.370934 versus control 0.359912: +0.01102, with image-bootstrap interval [+0.00481,+0.01682]. The stronger historical H9b calibration baseline was 0.462433, leaving a difference of -0.09150. The treatment produced 1,770 false positives versus H9b's 505. It was not promoted or submitted. A separate P59 training-only reliability gate admitted no additional calibration components and left that H9b baseline unchanged.

Later evaluation changes the interpretation of the favorable within-family calibration result. The original F1 experiment's 7,200-step control and treatment obtained 0.398796 and 0.394077 on 139 held-out TRAIN observations. P83 added three seeds; their matched 7,200-step treatment effects were -0.003721, -0.010176 and -0.015467.

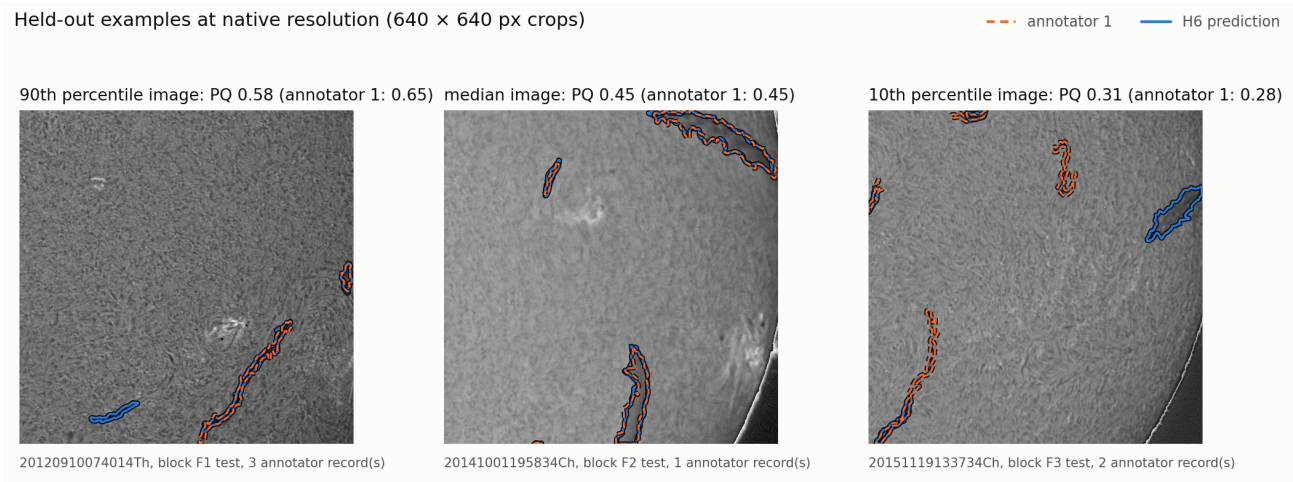


Figure 1: Measured H6 examples on held-out training observations, selected near the high, median and low image-quality quantiles. Orange outlines show one annotator and blue outlines show predictions. Image-level scores include the available annotation records; the illustrated annotator need not represent all evaluated contours. These are original observations and measured predictions.

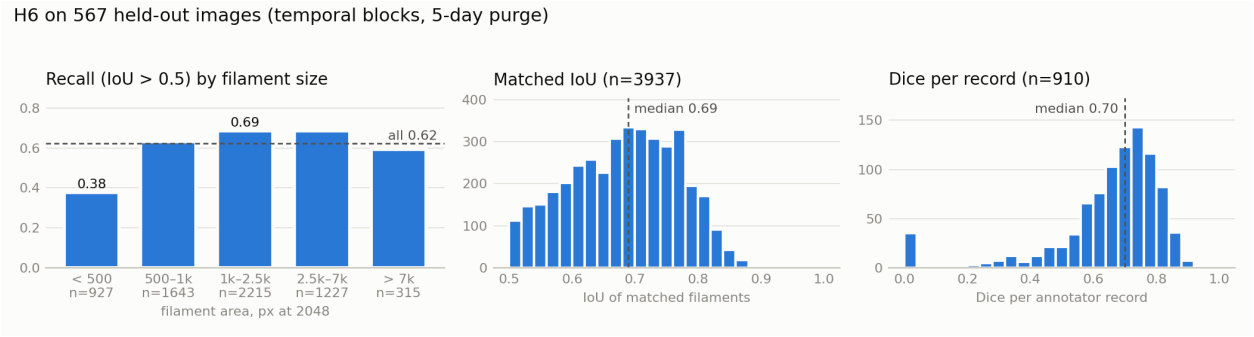


Figure 2: H6 diagnostics on 567 held-out training observations: recall by native filament area, IoU of matched objects, and foreground Dice per annotator record. The populations and metric definitions differ between panels and are shown in their labels. These measurements describe H6, not a new Kaggle evaluation of H9b.

The P84 cross-fold audit covered ten existing reports across F1–F4: three effects were positive and seven negative. All ten counting tuples independently reproduced their stored native PQ within 10^{-11}. Because folds and seeds reuse observations, these checks are descriptive correlated evidence, not ten independent tests. They support retaining H9b and avoiding another unsupported promotion [9].

Other retained controls include a deterministic dark-ridge system (local PQ 0.087–0.103) and a lower detector-confidence setting that added 94 true positives but 278 false positives on F4. These results show why increased foreground recall must be assessed alongside instance-level precision.

3.4 Reproduction, efficiency and release status

The public repository is Agnuxo1/solar-filament-segmentation-2026 [9]. It contains source, pinned dependencies, weight manifests and notebooks. Four original FP32 detector checkpoints are distributed as eight verified shards; the original detector values must not be confused with an earlier lossy FP16 export. Frozen H9b refinement uses the published original two checkpoints on CPU in FP32.

The strongest artifact-level result is P94: from the original no-site crop archives, public refinement and assembly code produced **1,241/1,241 identical RLE rows and the identical CSV SHA-256** as accepted submission 58112557. P93 independently corrected the identity of the reference file: an archive named as a submission was actually an intermediate. P95 subsequently reproduced the crop stage for 180/180 images when historical H3 and detector outputs were supplied. P99 independently reproduced detector consensus for all 180 images. Each statement has a separate bounded input contract.

The remaining upstream numerical differences matter to the full claim. In P102, independently regenerated public stages produced 14/15 identical RLE rows across a fixed four-image pilot;

component identities, scores and affine geometries matched, but 179 crop tensor elements differed. P103 completed the 180-image extension at 10:50:06 UTC on 13 November, inferring 176 additional images and reusing four pilot cases. It produced 1,241 rows: **1,101 position-wise exact RLE matches and 140 differences**, with identical complete predictions for 90/180 images. Instance counts matched for all 180 images; all followed `public_pipeline`, with zero historical fallbacks. The recorded CPU stage-coupling runtime was 1,661.308 s. The audit JSON and CSV SHA-256 digests begin `d67fe349750d6da9` and `8ae3160af9bddd5a`, respectively; the source package records both full hashes. These differences do not establish lower or higher segmentation accuracy, and no new PQ or Kaggle score is claimed.

P104 verified all ten weight hashes and static notebook interfaces without an end-to-end run. P103 starts from reconstructed stages and does not replace a clean execution of the revised portable notebook at commit `6843362`. P106 is being prepared on CUDA GPU 0 with public weights; no result is reported. The notebook audited at P104 contained a deterministic no-component fallback different from the historical control path. Until separately evaluated, regenerated output cannot inherit the accepted artifact's PQ 0.39.

Efficiency measurements are similarly scoped. P58 recorded 970.34 s for the 3,600 paired training steps and peak CUDA allocation 1.793 GiB on a 24-GiB RTX 3090. These are training measurements. A P75 inference batching pilot found gains below its preregistered +15% threshold and was not promoted. Neither result measures complete JPEG-to-CSV inference latency, energy, emissions or water use. Those quantities are not inferred from GPU utilization.

Remaining scientific limits include annotation disagreement, temporal correlation, small-object errors, repeated use of some local splits, partial leaderboard visibility and device-dependent quantization. The organizer form was acknowledged at 14:00:23 UTC (15:00:23 Madrid) on 13 November. Submission receipts and report revision provenance are recorded in the public repository [9]. A clean execution of the revised notebook remains a separate verification task.

4 Acknowledgment

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